

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTBS404 - PROBABILITY THEORY AND RANDOM PROCESSES

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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- 1 The addition theorem of probability is used when:
a) Events are mutually exclusive b) Events are independent
c) Events are exhaustive d) None of the above
- 2 If A and B are two independent events, then $P(A \cap B) = ?$
a) $P(A) + P(B)$ b) $P(A) \times P(B)$
c) $P(A) / P(B)$ d) $P(A) - P(B)$
- 3 Which distribution is appropriate for modeling the number of occurrences of an event in a fixed interval of time or space?
a) Binomial distribution b) Poisson distribution
c) Normal distribution d) Exponential distribution
- 4 The mathematical expectation of a discrete random variable X is given by:
a) $\int x f(x) dx$ b) $\sum x P(x)$
c) $\sum x^2 P(x)$ d) $\sum P(x)/x$
- 5 The variance of a random variable is:
a) $E(X)$ b) $E(X^2)$
c) $E[(X - \mu)^2]$ d) $\sqrt{E(X)}$
- 6 The correlation coefficient lies between:
a) 0 and 1 b) -1 and +1
c) $-\infty$ and $+\infty$ d) 0 and ∞
- 7 Spearman's rank correlation is used when:
a) Data is quantitative b) Data is qualitative
c) Data is ordinal d) Data is categorical
- 8 Which of the following best describes regression?
a) Finding the average b) Finding the relationship between variables
c) Testing a hypothesis d) Estimating probability
- 9 If the regression coefficient is 0, then:
a) There is a perfect positive relationship b) There is no linear relationship
c) There is a perfect negative relationship d) The correlation is 1

- 10 In regression analysis, the line that minimizes the sum of the squared deviations is known as:
- a) Mode line b) Best fit line
c) Standard line d) Mean line
- 11 Which is required to conduct a large sample test for proportion?
- a) Mean b) Median
c) Sample size > 30 d) Variance
- 12 Which of the following is a small sample test?
- a) Z-test b) Chi-square test
c) T-test d) F-test

Q2. Solve the following.

- A) Four cards are drawn from a pack of cards. Find the probability that (i) all the diamonds, (ii) there is one card of each suit, and (iii) there are two spades and two hearts. 6
- B) State and explain Bayes' theorem with an example 6

Q3. Solve the following.

- A) Define a random variable. Distinguish between discrete and continuous random variables with suitable examples. 6
- B) In a sample of 1000 cases the mean of a certain test is 14 and standard deviation is 2.5. Assuming the distribution to be normal, Find (i) How many students score between 12 and 15? (ii) How many students score above 18? [Given: $A(0.8) = 0.2881$, $A(0.4) = 0.1554$, $A(1.6) = 0.4452$] 6

Q4. Solve Any Two of the following.

- A) Describe Karl Pearson's correlation coefficient. Derive its formula and explain its significance. 6

B)

X marks in LA	23	28	42	17	26	35	29	37	16	46
Y marks in DM	25	22	38	21	27	39	24	32	18	44

For a group of students marks obtained by each student in the paper Linear Algebra(LA) and Discrete Mathematics (DM) are as follows. Calculate Karl Pearson's coefficient of correlation between x and y

C)

x	32	35	39	60	43	37	43	49	10	20
y	40	30	70	20	30	50	72	60	45	25

From the following data, calculate the rank correlation coefficient.

Q5. Solve Any Two of the following.

- A) The regression equation are: $-8x + 5y + 17 = 0$ and $-5x + 2y + 14 = 0$. If $\text{Var}(y) = 16$, Find (i) The mean values of x and y (ii) The correlation Co-efficient between x and y (iii) The standard deviation of x. 6
- B) Define linear regression. Derive the equation of the regression line of Y on X. Explain the meaning of regression coefficients. 6
- C) Obtain regression lines for the following data: 6

X	2	3	5	7	9	10	12	15
Y	2	5	8	10	12	14	15	16

Find (i) y when x = 6 and (ii) x when y = 20..

Q6. Solve Any Two of the following.

- A) What is mean by Testing of hypothesis, level of significance, critical region, Type-I and Type-II errors. 6
- B) A manufacturer claims that the mean breaking strength of safety belts for air 6 passengers produced in his factory is 1275 kg. A sample of 100 belts was tested and the mean breaking strength and Standard Deviation were found to be 1258 and 90 kg respectively. Test the manufacturer's claim at 1% Level of significance.
- C) In two large population there are 30% and 25% respectively of fair haired 6 peoples. Is this difference likely to be hidden in sample of 1200 and 900 respectively from the two populations? Use 5% level of significance.

*** End ***